

# Seeing with sound

When night falls, insect-eating bats take off from their daytime roosts and emerge to find food. Fluttering through the darkness on leathery wings, they catch prey in midair or snatch them from the ground. Unlike fruit-eating bats, they have tiny eyes, but they can swerve around all kinds of obstacles, from trees to phone lines, as they speed after their prey. The bats do this by using echolocation—a way of seeing their surroundings using rapid bursts of high-frequency sound. By listening for returning echoes, a bat builds up a picture of its world, including its food. Bats are not the only animals that use sound in this way. Several other kinds of mammal can echolocate, as can some birds that nest deep in caves.

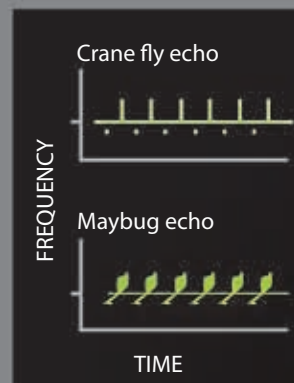


## Ground sound

Aside from shrews, tenrecs are the only land-based mammals known to use echolocation. Living in Africa and Madagascar, they feed at night and make clicking sounds with their tongues as they move around. Unlike a bat's pulses, a tenrec's clicks can be heard by human ears.

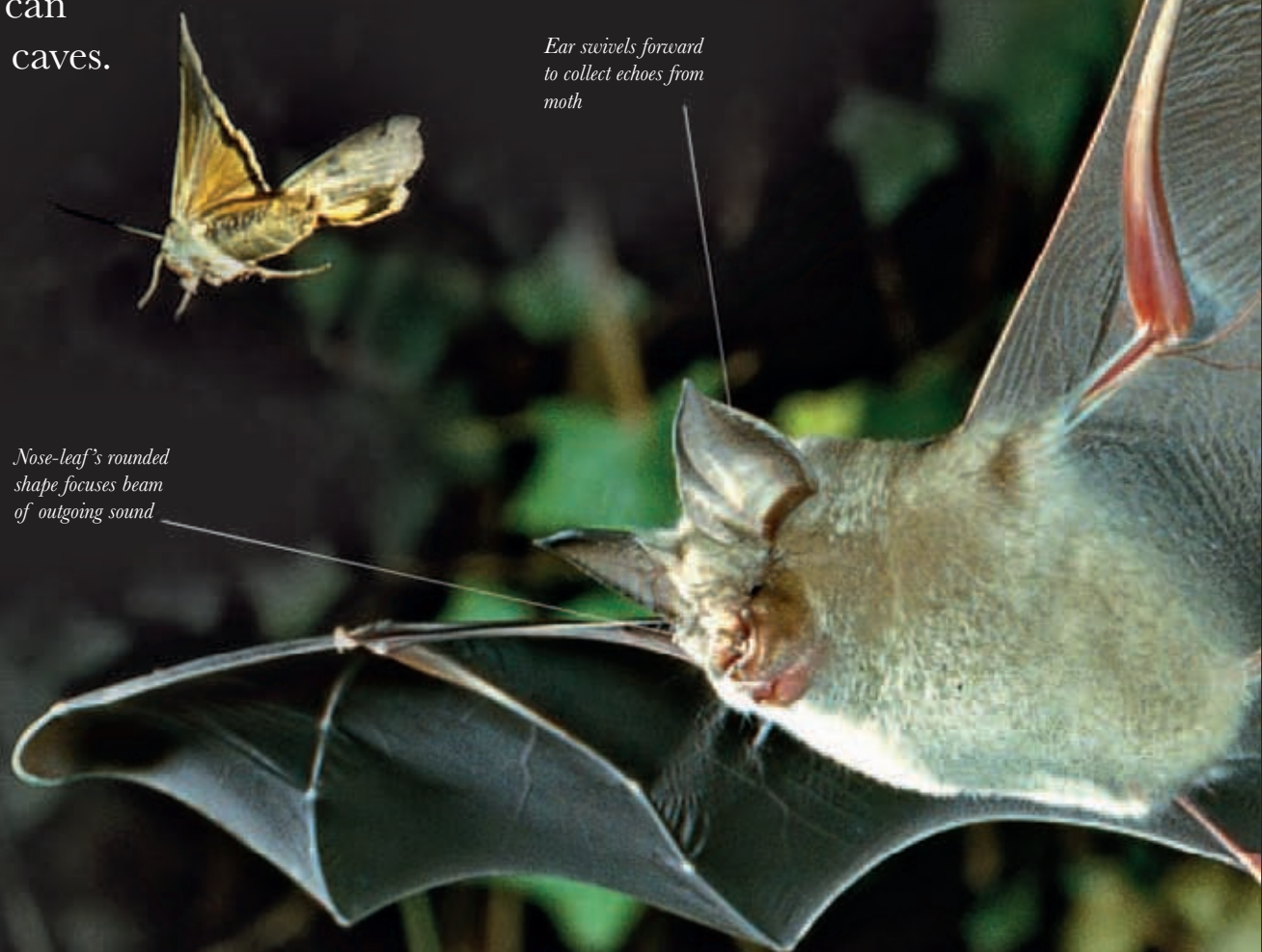
## Fly echoes

A hunting greater horseshoe bat squeaks with a constant pitch, or frequency. It distinguishes insect types by analyzing the echoes made by their beating wings. The echoes wobble in frequency due to Doppler shift, the effect that changes the pitch of a siren as it speeds past.



## Picture this

As it closes in on a moth, a greater horseshoe bat needs to know exactly how far away its prey is and how fast it is moving. It gives out more than 100 sound pulses a second—more than 20 times the rate when it is on patrol. It emits its sound pulses through its nostrils, in the center of its bizarre, horseshoe-shaped nose-leaf.



*Ear swivels forward to collect echoes from moth*

*Nose-leaf's rounded shape focuses beam of outgoing sound*